

Question number	Answer	Additional guidance	Mark
1(a)	Choose an item. an arrow going to the right	Accept correct arrow anywhere on the diagram ACCEPT arrow going to the left reject arrows in 2 different directions	(1)

Question number	Answer	Additional guidance	Mark
1(b)(i)	Choose an item. B (-70 mV) is the correct answer A is not the correct answer as the resting potential is not -90 mV is where membrane is hyperpolarised C is not the correct answer as the resting potential is not +48 mV D is not the correct answer as the resting potential is not +70 mV		(1)

Question number	Answer	Additional guidance	Mark
1(b)(ii)	Choose an item. C (W) is the correct answer A is not the correct answer as the membrane is not hyperpolarized at U. B is not the correct answer as the membrane is not hypopolarised at V. D is not the correct answer as the membrane is not hyperpolarised at X.		(1)

Question number	Answer	Additional guidance	Mark
1(b)(iii)	Choose an item. D (140 mV) is the correct answer A is not the correct answer as the difference is not 40 mV. B is not the correct answer as the difference is not 50 mV. C is not the correct answer as the difference is not 90 mV.		(1)

Question number	Answer	Additional guidance	
1(c)	An answer that includes the following points: • one correct difference in function (1) • second correct difference in function (1)	ACCEPT spinal cord for CNS NOT spine ACCEPT action potentials for impulses ACCEPT piece together from different sentences. Mix and match Reject messages / information / signals sensory neurones: sensory neurones carry (electrical) impulse towards the {CNS / spinal cord / brain / relay neurone / synapse} whereas.... sensory neurones {receive stimulation from receptors / sense organ} / sensory neurones synapse with receptors / sensory carry impulses (away) from {receptors / sense organ} sensory neurones release neurotransmitters (into synapse)} whereas.... Motor motor neurone {carry (electrical) impulse away / receives impulse} from {CNS / brain / spinal cord / relay neurone / synapse} motor neurone {carry electrical impulse towards / stimulate a response in / synapse with} {effector / muscle / gland / neuromuscular junction} motor neurones are stimulated by (relay) neurotransmitters	(2)

